Numerical Method

Assignment 1

1. What are the sources of errors? Discuss various types of errors.
2. Find one root of $x^3 -2x-5=0$ with the accuracy of 0.08% using bisection method.
3 Find at least one root of the equation $x^2+\tan x+e^x=0$ correct up to 3 significant digits using bisection method.
4. Find at least one root of the following equations using Newton Raphson method
   a. $e^{x-1} -5x^3$ correct up to 4 decimal places
   b. $x=e^{-x}$ correct up to 5 decimal places
5. Find a real root of following equations using secant method
   a. $f(x)=e^{\cos x} + \sin(x)$ up to 3 decimal places
   b. $f(x)= 3x+\sin x-e^x$ up to 5th iterations
6. Solve the non- linear equation $\log x - \cos x = 0$ by using false position method correct to four decimal places.

Hint: If two initial guesses of secant method brackets the root then it is called false position method. We use same method and formula of secant.

7. Find the root of $x-\sin[x]-(1/2) = 0$ correct to 4 decimal places using fixed point iteration method.
8. Find the first approximate root of the equation $\cos x = 3x - 1$ up to 4 decimal places.
9. Evaluate the polynomial $f(x) = x^4 + 3x^3 + 5x^2 + 7x + 9$ at $x = 2$ using Horner's Rule.